

AutoShotV2: Calibrated Classification and Post-Processing for Short-Video Shot Boundary Detection

Senh Manh Tang, Tien Manh Do, and Thai Son Tran

Abstract—Shot boundary detection (SBD) is a fundamental preprocessing step for video indexing, retrieval, summarization, and automated editing. Modern short-video platforms create a difficult operating regime: videos are brief, shots are extremely short, transitions are visually complex, and boundary frames are rare compared with non-boundary frames. This paper presents AutoShotV2, a lightweight improvement over the AutoShot architecture for short-video SBD. Instead of re-searching or retraining the full backbone, AutoShotV2 freezes the AutoShot feature extractor and trains a new two-branch classification head over 4864-dimensional cached features, exploring complementary one-hot and boundary-aware many-hot supervision. At inference time, the primary boundary logits are calibrated by temperature scaling and stabilized by one-dimensional Gaussian smoothing before thresholding. With a fixed deployment configuration ($T^* = 0.3878$), AutoShotV2 reaches an F1-score of 0.8545 on the SHOT test set of 200 short videos, improving over the reproduced AutoShot baseline by 1.40 F1 points and over the reported TransNetV2 baseline by 5.52 points, and transfers to BBC documentary footage (0.9656) while remaining limited on open-world ClipShots (0.7530). A controlled replication under a frozen validation-only protocol shows that the gain comes from calibrated post-processing rather than the training-loss alternatives, with paired bootstrap intervals supporting improvements on SHOT and ClipShots but not BBC. Temperature scaling reduces validation NLL from 0.013006 to 0.009591 and equal-width ECE from 0.007149 to 0.002214, while class-balanced ECE remains 0.070405, showing that aggregate frame-level calibration metrics are optimistic under severe boundary sparsity. A three-seed replication bounds training-seed F1 variability below 0.005 on every dataset. We release validation-selection manifests, per-video statistics, analysis code, and checksum-oriented artifact tooling.

Index Terms—Shot boundary detection, short video, AutoShot, frozen features, focal loss, probability calibration, temperature scaling, Gaussian smoothing, video understanding.

I. INTRODUCTION

Effectively parsing video content is a fundamental requirement for many downstream applications such as video indexing, content retrieval, summarization, and automated editing. Within this pipeline, shot boundary detection (SBD) serves as a critical preliminary step: it partitions a video into temporally coherent shots by identifying the exact frames where camera views, scene contents, or editing states transition [1], [2]. Classical methods based on hand-crafted cues such as color histograms and motion differences are efficient but fragile

under fast camera motion, flashes, and modern editing styles [3].

Deep architectures such as TransNet and TransNetV2 [5], [6] learn spatial-temporal representations directly from frame sequences and have significantly advanced the state-of-the-art. Concurrently, the rise of short-video platforms (e.g., TikTok, YouTube Shorts) has introduced a new, challenging operating regime: brief videos, dense editing, and heavy special effects. To address it, AutoShot [4] leverages neural architecture search (NAS) to find an optimal 3-D convolutional backbone, demonstrating strong performance on the newly introduced SHOT dataset. Further improving these models, however, encounters a severe bottleneck: the NAS pipeline requires immense computational overhead for supernet training and architecture evaluation. Moreover, short-video SBD suffers from extreme class imbalance — boundary frames are exceptionally rare — and complex visual effects that act as hard negatives.

Following the strategy of separating representation learning from task-specific adaptation, we study a more efficient refinement path for short-video SBD. Instead of re-running the expensive NAS process, we freeze the searched AutoShot backbone and improve only the decision stage, a method we term AutoShotV2. Its core component is a dual-branch classification head applied over the frozen 4864-dimensional cached features. To handle the class imbalance, we explore Focal Loss [9] on the primary branch and a boundary-aware many-hot supervision strategy on the auxiliary branch, providing denser temporal gradients without compromising the precise one-hot localization of the primary branch. Finally, because SBD decisions rely directly on thresholding model scores, we introduce a calibrated post-processing stage: temperature scaling [10] adjusts the numerical scale of the predictions without altering their ranking, and a one-dimensional Gaussian filter smooths out isolated false-positive peaks caused by sudden flashes or camera shakes.

With the fixed deployment configuration, AutoShotV2 achieves an F1-score of 0.8545 on the SHOT test set, improving over the reproduced AutoShot baseline and the reported TransNetV2 baseline by 1.40 and 5.52 F1 points, respectively. A controlled replication under a frozen, validation-only five-fold protocol disentangles the sources of this gain: the training-loss alternatives change F1 only marginally, whereas calibrated post-processing adds 1.62 F1 points on SHOT and 4.57 points on ClipShots over the BCE control, with paired bootstrap intervals confirming the deltas on SHOT and ClipShots but not BBC.

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In summary, our main contributions are:

- We propose AutoShotV2, an efficient SBD pipeline that improves short-video boundary detection without the computational burden of retraining the NAS-searched backbone, reaching 0.8545 F1 on SHOT with a fixed deployment configuration.
- We design a two-branch classification head and systematically explore Focal Loss and boundary-aware many-hot supervision as a training design space, showing through a controlled ablation that calibrated post-processing — not the more complex losses — drives the measured gain.
- We define a validation-only, source-stratified selection protocol that freezes temperature, smoothing, and threshold before any test evaluation, and quantify uncertainty with paired video bootstrap intervals and calibration diagnostics including adaptive and class-balanced ECE.
- We provide extensive experiments on SHOT, BBC, and ClipShots with strict cache identities, exact split/run manifests, a three-seed robustness study, and checksum-based artifact packaging.

The paper reports results on two tracks by design: the deployed checkpoint supplies the headline numbers and is preserved as tracked, checksummed result files, while a controlled replication with full logit caches anchors every confidence interval, calibration diagnostic, and ablation (Section V-B).

II. RELATED WORK

Shot boundary detection. Shot boundary detection has long been an active research area in video processing. Early systems primarily measured frame-to-frame discontinuities using hand-crafted visual features. Methods relying on histogram differences, edge-change ratios, motion compensation, and thresholding-based heuristics are computationally simple. However, they are prone to mistaking lighting changes, camera motion, or object movement for actual shot boundaries [3]. Comprehensive survey studies categorize the major challenges in this domain into abrupt cuts, gradual transitions, illumination changes, object/camera motion, and dataset-dependent evaluation protocols [1], [2].

Deep SBD models. Following the success of deep learning in vision tasks, deep models were introduced to learn temporal patterns that are difficult to encode manually [13], [14], from hierarchical clustering for broadcast re-use [20] to large-scale spatio-temporal CNNs [21]. TransNet models SBD as a dense frame-level prediction task over a sequence of low-resolution frames [6]. TransNetV2 further improves this concept by incorporating dilated 3-D convolutions, frame-similarity features, color histograms, and dual prediction heads for both single-frame and all-transition labels [5]. These models offer a fast and accurate inference pipeline, establishing them as strong baselines for practical SBD applications.

AutoShot and short-video SBD. Most existing datasets, such as BBC, RAI, and ClipShots, mainly represent documentaries, broadcast content, or longer user-generated videos. Recognizing this gap, AutoShot introduces the SHOT dataset specifically to capture short-video characteristics: brief duration, dense editing, vertical layouts, complex gradual transitions, and heavy special effects [4]. AutoShot leverages NAS

[8], [11], [12] to search over approximately 83.9 million candidate architectures composed of 3-D separable convolution blocks and temporal Transformer blocks [22], training a single-path one-shot supernet [23] explored with Bayesian optimization rather than differentiable relaxation [24]. While the searched backbone yields notable F1 improvements on SHOT, the computational cost of the search and retraining phases remains prohibitively high.

Imbalance, calibration, and temporal smoothing. Frame-level SBD is severely imbalanced because boundary frames are rare. Focal Loss reduces the contribution of easy examples and is therefore a plausible alternative to BCE [9], while soft and neighborhood-expanded targets such as label smoothing [25] motivate the boundary-aware many-hot supervision we explore. SBD decisions are also threshold-sensitive, making score calibration and temporal stability important. Temperature scaling rescales logits without changing their ranking [10]; classical alternatives include Platt scaling [15] and isotonic regression [16], which fit richer mappings at the cost of more calibration data — a poor fit for our small validation pool. Measurement itself is subtle: equal-width ECE can be dominated by the majority class, motivating adaptive and class-conditional estimators [17], which is why we report equal-mass and class-balanced variants alongside ECE. Temporal smoothing suppresses isolated peaks. We evaluate these mechanisms separately rather than assuming that a more complex training loss is necessary.

Efficient adaptation of frozen representations. Reusing a pretrained backbone without updating it is a long-standing strategy, from linear probing of frozen features to parameter-efficient modules such as adapters [18] and low-rank updates [19]. Our setting is an extreme point on this spectrum: the searched AutoShot representation is fully frozen, a small classification head is retrained on cached features, and all remaining adaptation happens in calibrated post-processing. This keeps the adaptation cost at CPU scale and makes every downstream decision auditable, at the price of forgoing representation-level domain adaptation — a trade-off our ClipShots results make explicit.

III. SBD FOUNDATIONS: TRANSNETV2 AND AUTOSHOT

This section formulates the shot boundary detection problem and reviews the architectural components of the TransNetV2 family and the AutoShot search that produced the backbone AutoShotV2 builds upon.

A. Problem Formulation

Given a video represented as an ordered sequence of T frames $\{x_t\}_{t=1}^T$, SBD predicts a binary sequence $\{b_t\}_{t=1}^T$, where $b_t=1$ denotes a shot boundary (an abrupt cut or a gradual transition) at frame t . A model outputs a boundary probability sequence $\{p_t\}_{t=1}^T$ with $p_t \in [0, 1]$, and the final boundary set is obtained by thresholding:

$$\hat{b}_t = \begin{cases} 1, & p_t > \theta, \\ 0, & p_t \leq \theta, \end{cases} \quad (1)$$

where θ is selected on validation data. Following standard practice, a predicted boundary counts as correct when it falls within 2 frames of a ground-truth boundary, and consecutive positive predictions are grouped into one transition.

B. Spatial-Temporal Factorized Convolution

Processing raw frame sequences with full 3-D convolutional kernels of size $k \times k \times k$ is expensive. The TransNetV2 family therefore factorizes each 3-D convolution into a 2-D spatial convolution ($1 \times k \times k$, learning edges, textures, and color regions per frame) followed by a 1-D temporal convolution ($k \times 1 \times 1$, capturing motion patterns along the temporal axis). With C input/output channels and $k=3$, the parameter count drops from $27C^2$ to $9C^2 + 3C^2 = 12C^2$, a reduction of roughly 56% that also imposes a useful inductive bias separating spatial content from temporal motion.

C. Dilated DDCNN Blocks

The backbone stacks Dilated Convolutional Neural Network (DDCNN) blocks. Each block splits its input features into four parallel temporal-convolution branches with exponentially increasing dilation rates $d \in \{1, 2, 4, 8\}$, concatenates the branch outputs along the channel dimension, and applies batch normalization with ReLU activation. For temporal kernel size $K=3$, the cumulative receptive field after block k grows as

$$R_k = R_{k-1} + 2(K-1) \sum_{d \in \{1, 2, 4, 8\}} d = R_{k-1} + 60, \quad (2)$$

with $R_0=1$. Six stacked blocks yield a theoretical receptive field of up to 361 frames; since inference uses sliding windows of 100 frames, the effective context is capped by the window but remains wide enough for long dissolves and fades.

D. Similarity Branch

A parallel similarity branch complements the convolutional features with two forms of visual consistency: (i) a 512-bin RGB histogram per frame, whose cosine distances between consecutive frames yield a 101-dimensional color-shift descriptor, and (ii) learned similarity, the cosine similarity between normalized deep features from intermediate convolutional layers. Both are fused through a fully connected layer with 128 units and ReLU activation, and the result is concatenated with the convolutional features before classification.

E. AutoShot Backbone Search

While TransNetV2 fixes a configuration of identical DDCNN blocks, AutoShot searches the backbone configuration for short-form video with neural architecture search [4], [8], [11], [12]. The space covers four DDCNN block variants (differing in channel configuration and spatial feature sharing) across six layers plus a choice of zero to four Transformer layers [22] at the seventh, about 83.9×10^6 candidate architectures in total. The search trains a single-path one-shot supernet [23] explored with Bayesian optimization (differentiable alternatives such as DARTS [24] were considered by the original authors), and the selected model is refined with knowledge

distillation and weight grafting. The architecture found under the F1 criterion uses DDCNNV2 blocks of 4–12 filters with shared spatial convolutions in layers 2–4 and no Transformer layer. Its penultimate stage exposes a 4864-dimensional feature vector, which we freeze and cache:

$$h_t \in \mathbb{R}^{4864}. \quad (3)$$

All AutoShotV2 components operate on h_t without updating the backbone.

IV. PROPOSED AUTOSHOTV2

AutoShotV2 improves the decision stage of SBD: it trains a new classification head on the frozen features of Eq. (3) and adds probability calibration and temporal smoothing before thresholding. Fig. 1 shows the deployed inference pipeline.

A. Classification Head Architecture

The head maps each cached feature vector h_t to a shared hidden state

$$u_t = \text{Dropout}(\text{ReLU}(W_f h_t + c_f)), \quad (4)$$

with $W_f \in \mathbb{R}^{1024 \times 4864}$ and dropout probability 0.5. Two parallel linear branches then produce logits

$$z_t^{(1)} = W_1 u_t + c_1, \quad z_t^{(2)} = W_2 u_t + c_2. \quad (5)$$

The primary branch $z_t^{(1)}$ is trained against the exact one-hot boundary labels and is the only branch used at inference. The auxiliary branch $z_t^{(2)}$ can be supervised with boundary-aware many-hot targets to supply denser gradients around each transition. Fig. 2 illustrates the head.

B. Training Objectives as a Design Space

Boundary frames are exceptionally rare ($< 1\%$ of frames), so the training loss is a natural design axis. The baseline objective is binary cross-entropy on the primary branch,

$$\mathcal{L}_{\text{BCE}} = -y_t \log \sigma(z_t^{(1)}) - (1 - y_t) \log(1 - \sigma(z_t^{(1)})). \quad (6)$$

We explore two alternatives. First, Focal Loss [9] down-weights easy negatives,

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t^*)^\gamma \log(p_t^*), \quad (7)$$

where $p_t^* = p_t$ for a positive target and $1 - p_t$ otherwise, and $\alpha_t = \alpha y_t + (1 - \alpha)(1 - y_t)$; grid search on validation selects $\gamma = 1.0$ and $\alpha = 0.5$. Second, boundary-aware many-hot supervision [25] expands each ground-truth boundary t_b to its ± 1 frame neighborhood,

$$y_{\text{mh}}[t] = \begin{cases} 1, & \exists t_b \text{ s.t. } |t - t_b| \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

and supervises the auxiliary branch with weight 0.3: $\mathcal{L}_{\text{total}} = \mathcal{L}(\hat{y}^{(1)}, y_{\text{oh}}) + 0.3 \mathcal{L}(\hat{y}^{(2)}, y_{\text{mh}})$. The small auxiliary weight prevents the dense targets from blurring the primary branch's localization.

These objectives are evaluated as a design space rather than assumed beneficial: the controlled ablation of Section V-F shows that plain BCE matches both alternatives once post-processing is applied, so the analytic track of this paper anchors on the BCE configuration.

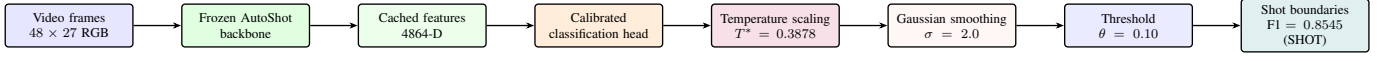


Fig. 1. AutoShotV2 inference pipeline. The searched AutoShot backbone is frozen; AutoShotV2 retrains the decision stage and adds calibration and temporal smoothing before thresholding. The controlled replication of Section IV-D re-fits its own temperature on its validation pool under the frozen validation-only protocol.

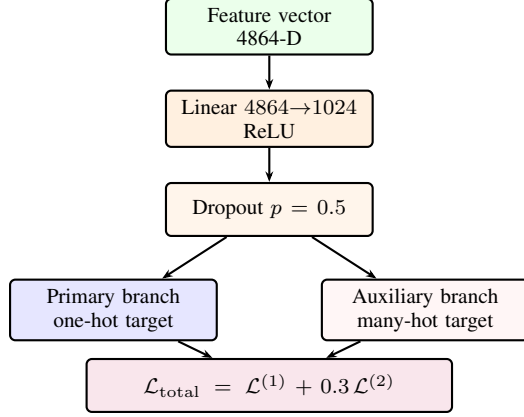


Fig. 2. AutoShotV2 classification head. The primary branch preserves sharp boundary localization; the auxiliary branch can supply denser supervision around each boundary. The controlled ablation selects plain BCE on the primary branch.

C. Probability Calibration and Temporal Smoothing

SBD decisions rely directly on thresholding model scores, so probability quality matters as much as ranking. Temperature scaling [10] rescales the primary logit without changing its ranking:

$$p_t^{\text{cal}} = \sigma\left(\frac{z_t^{(1)}}{T}\right), \quad T^* = \arg \min_T \mathcal{L}_{\text{BCE}}\left(\sigma\left(\frac{z_t^{(1)}}{T}\right), y\right), \quad (9)$$

with T^* fitted on validation data. Calibration cannot remove isolated temporal peaks caused by flashes or camera shakes, so a one-dimensional Gaussian filter smooths the calibrated sequence:

$$\tilde{p}_t = \sum_k G_\sigma(k) p_{t-k}^{\text{cal}}, \quad G_\sigma(k) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{k^2}{2\sigma^2}\right). \quad (10)$$

The deployed checkpoint fits $T^* = 0.3878$ on its validation pool; the controlled replication re-fits its own temperature under the frozen protocol below. Both use $\sigma = 2.0$ and $\theta = 0.10$.

D. Validation-Only Protocol Selection

For the controlled track, we select the complete operating point using five-fold validation. Folds are stratified by source dataset. For each held-out fold, temperature is fitted on the other four folds by minimizing binary cross-entropy. We search $\sigma \in \{0, 1, 2, 3\}$ and $\theta \in \{0.05, 0.10, \dots, 0.50\}$. Each pair is scored by first computing F1 per source dataset within each held-out fold, averaging those per-source values into a fold-level macro-F1, and then averaging the macro-F1 over the five folds. Ties prefer lower smoothing and then the threshold

TABLE I
DATASET CHARACTERISTICS USED FOR SBD EVALUATION

Property	BBC	ClipShots	SHOT
Content type	Documentary	Open-world video	Short video
Total videos	11	4,039	853
Evaluation videos	11	500	200
Avg. video length	2,945 s	237 s	39.5 s
Avg. shot length	6.57 s	15.34 s	2.59 s
Hard cuts	Many	128,000+	8,890
Gradual transitions	Many	38,000+	2,716
Main challenge	Gradual trans.	Diverse effects	Fast editing

nearest 0.10. After selecting σ and θ , the final temperature is fitted on all validation videos; the selected values are listed in Table IV. The manifest containing folds, key hashes, search space, and selected values is written before test evaluation. We denote the resulting BCE plus temperature plus Gaussian configuration B4.

E. Cache and Run Provenance

The revised pipeline separates a fixed data seed from the training seed. Before feature extraction, it deterministically selects the training-video list and hashes that exact list. A sample cache is reusable only when its schema version, metadata hash, base-checkpoint hash, selected-key hash, video budget, sampling parameters, boundary width, and data seed all match. Each run writes the selected video IDs, per-video sample counts, skipped inputs, configuration, checkpoint hash, and logit-key hash. This strict identity replaces the earlier path-tolerant cache check that could not prove which 400 videos produced the historical cache.

V. EXPERIMENTS

A. Datasets and Training Setup

We evaluate on the 200-video SHOT test split [4], all 11 BBC Planet Earth videos [20], and the 500-video ClipShots test split [7]. SHOT is the primary short-video benchmark; BBC tests long documentary footage with many smooth gradual transitions; ClipShots tests open-world transfer. Table I summarizes their characteristics.

AutoShotV2 trains only the classification head with Adam at a constant learning rate; Table II lists the hyperparameters and how each was selected. Performance is measured by precision, recall, and F1, micro-averaged from transition-level TP/FP/FN counts with a matching tolerance of ± 2 frames.

B. Evaluation Tracks and Reproducibility Tiers

The paper reports two evaluation tracks with different reproducibility tiers. *Track 1 (deployed checkpoint)* provides the headline numbers: the head trained for deployment with the

TABLE II
AUTO SHOT V2 HYPERPARAMETERS

Component	Value	Selection
Focal Loss γ	1.0	Grid search
Focal Loss α	0.5	Grid search
Learning rate	10^{-5}	Adam, constant
Weight decay	10^{-4}	Fixed
Batch size	512	Frame-level
Maximum epochs	20	Fixed budget
Dropout	0.5	Fixed
Many-hot weight	0.3	Validation
Gaussian σ	2.0	Validation
Deploy temperature T^*	0.3878	Validation BCE

TABLE III
F1-SCORE COMPARISON ACROSS EVALUATION DATASETS

Method	SHOT	BBC	ClipShots
AutoShot, reported in original paper	0.8410	0.9710	0.7870
AutoShot, reproduced	0.8405	0.9554	0.7753
TransNetV2, reported baseline	0.7993	0.9620	0.7760
TransNetV2, reproduced	—	0.9593	0.7454
AutoShot reproduced + Gaussian smoothing	0.8480	0.9510	0.7987
AutoShotV2 (ours), fixed deploy threshold	0.8545	0.9656	0.7530

fixed configuration $T^* = 0.3878$, $\sigma = 2.0$, $\theta = 0.10$. Its per-dataset result JSON files are tracked and checksummed, but its test logits were not retained, so logit-level analyses cannot be re-run for this checkpoint. *Track 2 (controlled replication)* retrains the head under the frozen validation-only protocol of Section IV-D (configuration B4) with full logit caches; every paired bootstrap, calibration diagnostic, ablation, and seed study in this paper is anchored on this track. Because the replication was trained in a separate environment with its own validation pool, its absolute scores differ slightly from the deployed checkpoint; to keep a single consistent set of absolute numbers, all replication results are therefore reported as *differences* against its internal BCE control (A1), which are invariant to that environment offset, while absolute F1 values in this paper always refer to the deployed checkpoint.

C. Main Results

Table III reports F1 across SHOT, BBC, and ClipShots. With the fixed deployment configuration, AutoShotV2 reaches 0.8545 on SHOT and 0.9656 on BBC, improving over the reproduced AutoShot baseline by 1.40 F1 points and over the reported TransNetV2 baseline by 5.52 points on SHOT (Fig. 3). On ClipShots the fixed configuration reaches 0.7530; a separate test-set threshold sweep reaches 0.7557 at $\theta = 0.15$, reported for transparency but not treated as the fixed deployment result. Reported baselines use their authors' protocols, whereas reproduced baselines use our evaluation pipeline; the TransNetV2 reproduction was not evaluated on SHOT.

D. Controlled Component Analysis

All remaining analyses use Track 2: the controlled B4 replication whose protocol is summarized in Table IV.

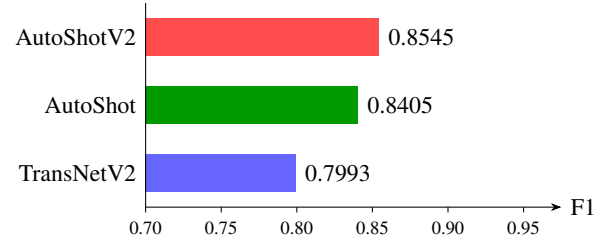


Fig. 3. F1-score comparison on SHOT. AutoShotV2 improves the reproduced AutoShot baseline by 1.40 points and the reported TransNetV2 baseline by 5.52 points.

TABLE IV
FIXED REPRODUCIBLE EVALUATION PROTOCOL (CONTROLLED TRACK)

Item	Setting
Training objective	BCE, one-hot labels
Training / data seed	42 / 42
Training budget	20 epochs, batch size 512
Sampling	At most 160 frames/video, 3 negatives/positive
Validation size	200 videos
Validation selection	5 source-stratified folds
Selection criterion	Mean macro-F1 across sources/folds
Selected temperature	0.6618
Gaussian smoothing	$\sigma = 2.0$
Decision threshold	$\theta = 0.10$
Matching tolerance	± 2 frames
Bootstrap	10,000 video resamples, seed 42

TABLE V
PAIRED VIDEO BOOTSTRAP DIFFERENCE, B4 MINUS A1

Dataset	$\Delta F1$	95% CI	Excl. 0
SHOT	0.0162	[0.0093, 0.0236]	Yes
BBC	-0.0000	[-0.0033, 0.0041]	No
ClipShots	0.0457	[0.0267, 0.0670]	Yes

Relative to the A1 control, the selected B4 configuration improves SHOT by 1.62 F1 points and ClipShots by 4.57 points, while BBC is unchanged after rounding. Temperature scaling alone preserves the validation-selected operating point; Gaussian smoothing is the component that changes boundary decisions.

Table V bootstraps the per-video difference between A1 at its validation-selected threshold and B4 at its frozen operating point (10,000 paired video resamples). The intervals support a positive B4 delta on SHOT and ClipShots. The BBC interval contains zero, so we make no improvement claim for that dataset; BBC also has only 11 sampling units, so its interval should be interpreted cautiously.

E. Calibration Diagnostics

We evaluate frame probabilities on the combined 200-video validation split using negative log-likelihood (NLL), Brier score, expected calibration error with 15 equal-width

TABLE VI
FRAME-LEVEL VALIDATION CALIBRATION DIAGNOSTICS

Variant	NLL	Brier	ECE	AECE	B-ECE
Before temperature scaling	0.013006	0.002194	0.007149	0.007149	0.080506
After temperature scaling	0.009591	0.001911	0.002214	0.002243	0.070405

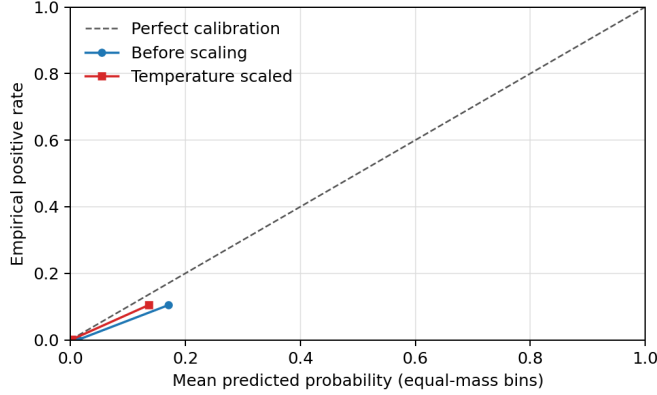


Fig. 4. Reliability diagram on the combined validation split. Temperature scaling reduces aggregate error. Equal-mass bins expose sparsely populated confidence regions.

TABLE VII
TEST CALIBRATION AT THE VALIDATION-SELECTED TEMPERATURE

Dataset	NLL	Brier	ECE	AECE	B-ECE
SHOT	0.021689	0.004808	0.003014	0.002844	0.133525
BBC	0.013305	0.001662	0.000851	0.001651	0.092471
ClipShots	0.007654	0.001565	0.002520	0.002521	0.156939

bins (ECE), equal-mass adaptive ECE (AECE), and class-balanced ECE (B-ECE). Table VI shows that temperature scaling improves all diagnostics. The much larger B-ECE reveals that aggregate frame metrics are dominated by negative frames. Validation values remain diagnostic because the same split fits temperature.

Table VII reports untouched-test calibration using the temperature selected on validation. ClipShots has the largest B-ECE despite a small aggregate ECE, consistent with domain-shifted false positives.

F. Controlled Training Ablation

Table VIII reports the outcome of the design space of Section IV-B as F1 differences against the A1 control. A2, A3, and B1 show that Focal Loss and many-hot supervision do not improve the control. B5 combines these training changes with B4 post-processing and is numerically similar on SHOT but slightly weaker on BBC and ClipShots. The gain therefore comes from calibrated post-processing rather than the more complex training objective. A0 is omitted because its per-dataset best-threshold protocol is not a direct controlled comparison.

TABLE VIII
CONTROLLED PHASE 2 ABLATION: $\Delta F1$ AGAINST THE A1 BCE CONTROL

Configuration	SHOT	BBC	ClipShots
A2 – Focal only	-0.0000	-0.0003	-0.0017
A3 – Many-hot only	-0.0003	-0.0008	+0.0021
P1 – Gaussian only	+0.0054	-0.0134	+0.0536
P2 – Temperature only	+0.0000	+0.0000	+0.0000
B1 – Focal + many-hot	+0.0006	-0.0011	+0.0023
B4 – Temperature + Gaussian (selected)	+0.0162	-0.0000	+0.0457
B5 – Full candidate	+0.0164	-0.0019	+0.0425

TABLE IX
EFFICIENCY AND RUN-ARTIFACT STATISTICS

Item	Value
Frozen backbone parameters	14,299,202
Trainable Phase 2 head parameters	4,983,810
Deployed primary-path parameters	4,982,785
Training videos / sampled frames	400 / 33,174
Training epochs / seed	20 / 42
Post-process CPU cost	21.1 ms / 10^6 frames

TABLE X
CLIPSHOTS RECALL BY GROUND-TRUTH TRANSITION TYPE

Type	GT	Matched	Missed	Recall
Cut	4907	4412	495	0.8991
Gradual	2302	1862	440	0.8089

G. Efficiency and Reproducibility

The backbone remains frozen, so only the Phase 2 head is optimized. Table IX reports parameter and run-artifact statistics. The CPU post-processing measurement covers temperature scaling and Gaussian smoothing on cached logits. End-to-end video decoding and backbone inference are excluded. The original run artifact did not retain elapsed training time, so we do not estimate or reconstruct it.

H. ClipShots Transition-Type Recall

We regenerate the ClipShots breakdown directly from the B4 logits and `test.json`. One malformed annotation with reversed endpoints is normalized by sorting the pair. A ground-truth interval is a cut when its width is at most one frame and gradual otherwise. Predictions are class-agnostic, so they are matched once and recall is grouped by the matched ground-truth type. Type-specific precision and F1 are not reported because the model does not predict a transition type.

The gradual-transition recall of 0.8089 is lower than the cut recall of 0.8991, but it is not the dominant source of the overall ClipShots gap. The aggregate ClipShots precision is only 0.6662, indicating that open-world effects and domain shift create many false-positive peaks.

I. Training-Seed Robustness

To quantify optimization variability, we retrained the BCE one-hot head 3 times with training seeds 42, 43, and 44 while

holding the data seed, the training-video selection, and the sample cache identity fixed. Each run selected its own post-processing configuration with the frozen five-fold validation-only protocol of Section IV-D before any test evaluation. Consistent with the delta-only reporting of Section V-B, we summarize this study by its dispersion: the sample standard deviation of test F1 across seeds is 0.0007 on SHOT, 0.0008 on BBC, and 0.0049 on ClipShots — below 0.005 on every dataset and an order of magnitude smaller than the paired bootstrap interval widths of Table V. The pipeline is therefore stable with respect to head-training randomness; this study isolates training-seed sensitivity end to end, complementing the component analysis of Table VIII.

J. Limitations

The study has four important limitations. First, the headline numbers come from the deployed checkpoint, whose result JSON files are tracked and checksummed but whose test logits were not retained (Section V-B); the controlled track therefore carries every logit-level claim, and the three-seed study of Section V-I bounds optimization variability (F1 sample standard deviation at or below 0.005 on every dataset) without reproducing either checkpoint bit for bit. Second, the historical sample cache recorded 400 videos and 33,174 frames but not the exact selected IDs; strict manifests fix this for future runs but cannot retroactively prove the old cache contents. Third, validation calibration diagnostics reuse the temperature-selection data. Fourth, the frozen representation does not fully control cross-domain false positives on ClipShots.

VI. DISCUSSION

Where the gain comes from. The controlled ablation resolves the design space of Section IV-B: training-side alternatives (Focal Loss, many-hot supervision) leave the BCE control statistically unchanged, whereas post-processing moves the operating point — Gaussian smoothing changes boundary decisions and temperature scaling improves probability quality without changing the ranking. For frozen-feature pipelines this ordering matters practically, because the post-processing search costs CPU milliseconds per million frames while re-training alternatives cost GPU hours, and because the search can be re-run per deployment domain without touching the representation.

What aggregate calibration hides. After temperature scaling the equal-width ECE falls to 0.002214, yet the class-balanced ECE remains 0.070405 — roughly thirty times larger. The gap is a direct consequence of boundary sparsity: non-boundary frames dominate every confidence bin, so a near-perfect aggregate score is compatible with poorly calibrated probabilities exactly on the rare positive class that the task cares about. Reporting only aggregate ECE would therefore overstate calibration quality for SBD and, we suspect, for other sparse temporal-event tasks. Class-wise or event-conditional calibration (for example, fitting a separate temperature on boundary-adjacent frames) is a natural next step, but it requires an independent calibration split that the present protocol does not reserve.

Sources of uncertainty. The two uncertainty analyses answer complementary questions. Paired video bootstrap intervals (Table V) quantify how much the measured deltas depend on which videos were sampled, and remain the dominant term: their widths are an order of magnitude larger than the seed-induced standard deviations of Section V-I. Training-seed variability, the usual objection to single-run reports, is bounded below 0.005 F1 across all three datasets by the three-seed study. The remaining unquantified term is protocol variability — the frozen five-fold selection is itself a function of the validation videos — which could be probed by repeating the selection over bootstrap-resampled validation sets.

The ClipShots gap is a precision problem. The transition-type breakdown shows gradual-transition recall (0.8089) trailing cut recall (0.8991), but the dominant loss is aggregate precision (0.6662): open-world content produces false-positive peaks that survive smoothing. A frozen representation cannot adapt to this distribution shift by construction, so threshold or temperature adjustments only trade recall for precision along a fixed curve. Closing the gap likely requires domain-adaptive components — either lightweight adapters on the frozen backbone or domain-specific post-processing selection — rather than further tuning of the present search space.

VII. CONCLUSION

We presented AutoShotV2, a compact refinement framework for short-video shot boundary detection. By freezing the NAS-searched AutoShot backbone and retraining a calibrated two-branch classification head, the method addresses the extreme class imbalance characteristic of short videos without repeating the computationally expensive architecture search. Temperature scaling and Gaussian smoothing provide the calibrated decision pipeline used at deployment, reaching F1 scores of 0.8545 on SHOT, 0.9656 on BBC, and 0.7530 on ClipShots with a single fixed configuration.

The controlled replication clarifies where the improvement comes from: Focal Loss and many-hot supervision leave the BCE control statistically unchanged, whereas calibrated post-processing yields paired-bootstrap gains on SHOT and ClipShots; a three-seed study bounds training-seed F1 variability below 0.005 on every dataset, and class-balanced calibration error shows why very small aggregate ECE values should not be read as uniformly calibrated boundary probabilities. Every reported number is regenerated and checked from tracked artifacts with exact cache identities and frozen selection manifests.

Cross-domain analysis identifies the main open problem: on open-world ClipShots the dominant loss is precision, as visual effects and domain shift create false-positive peaks that survive smoothing. Future work should reserve an independent calibration split, extend calibration to boundary-conditional estimates, and explore domain-adaptive components — lightweight adapters on the frozen backbone or per-domain post-processing selection — together with multimodal cues such as audio for robustness where visual evidence alone is ambiguous.

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REFERENCES

- [1] H. Seridi, M. Kurulay, Z. Kouahla, B. Farou, M. N. Kouahla, M. A. Ferrag, and M. A. Boudia, "Shot boundary detection: Fundamental concepts and survey," in *CEUR Workshop Proceedings*, vol. 2589, 2020.
- [2] T. Kar, P. Kanungo, S. N. Mohanty, S. Groppe, and J. Groppe, "Video shot-boundary detection: Issues, challenges and solutions," *Artificial Intelligence Review*, vol. 57, no. 104, 2024.
- [3] S. H. Abdulhussain, "Methods and challenges in shot boundary detection," *Entropy*, vol. 20, no. 4, p. 214, 2018.
- [4] W. Zhu, Y. Yu, W. Huang, J. Huang, W. Feng, X. Wang, X. Yang, C. Hu, C. Liu, B. Fan, and T. He, "AutoShot: A short video dataset and state-of-the-art shot boundary detection," in *Proc. IEEE/CVF CVPR Workshops*, 2023.
- [5] T. Soucek and J. Moravec, "TransNet V2: An effective deep network architecture for fast shot transition detection," *arXiv preprint arXiv:2008.04838*, 2020.
- [6] T. Soucek, J. Moravec, and J. Lokoc, "TransNet: A deep network for fast detection of common shot transitions," *arXiv preprint arXiv:1906.03363*, 2019.
- [7] S. Tang, L. Feng, Z. Kuang, Y. Chen, and W. Zhang, "Fast video shot transition localization with deep structured models," *arXiv preprint arXiv:1808.04234*, 2018.
- [8] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," *Journal of Machine Learning Research*, vol. 20, no. 55, pp. 1–21, 2019.
- [9] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. International Conference on Machine Learning*, 2017, pp. 1321–1330.
- [11] C. White, M. Safari, R. Sukthanker, B. Ru, T. Elsken, A. Zela, D. Dey, and F. Hutter, "Neural architecture search: Insights from 1000 papers," *arXiv preprint arXiv:2301.08727*, 2023.
- [12] P. Ren, Y. Xiao, X. Chang, P.-Y. Huang, Z. Li, X. Chen, and X. Wang, "A comprehensive survey of neural architecture search: Challenges and solutions," *arXiv preprint arXiv:2006.02903*, 2020.
- [13] A. Esteve et al., "Shot boundary detection with 3D depthwise convolutions and visual attention," *Sensors*, vol. 23, no. 12, p. 5620, 2023.
- [14] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE International Conference on Computer Vision*, 2015, pp. 4489–4497.
- [15] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*. Cambridge, MA, USA: MIT Press, 1999, pp. 61–74.
- [16] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.
- [17] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, "Measuring calibration in deep learning," in *Proc. IEEE/CVF CVPR Workshops*, 2019.
- [18] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *Proc. International Conference on Machine Learning*, 2019, pp. 2790–2799.
- [19] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. International Conference on Learning Representations*, 2022.
- [20] L. Baraldi, C. Grana, and R. Cucchiara, "Shot and scene detection via hierarchical clustering for re-using broadcast video," in *Proc. Computer Analysis of Images and Patterns*, 2015, pp. 801–811.
- [21] A. Hassani, M. Elgharib, A. Selim, S.-H. Bae, M. Hefeeda, and W. Matusik, "Large-scale, fast and accurate shot boundary detection through spatio-temporal convolutional neural networks," *arXiv preprint arXiv:1705.03281*, 2017.
- [22] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [23] Z. Guo, X. Zhang, H. Mu, W. Heng, Z. Liu, Y. Wei, and J. Sun, "Single path one-shot neural architecture search with uniform sampling," in *Proc. European Conference on Computer Vision*, 2020, pp. 544–560.
- [24] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," in *Proc. International Conference on Learning Representations*, 2019.
- [25] R. Müller, S. Kornblith, and G. E. Hinton, "When does label smoothing help?" in *Proc. Advances in Neural Information Processing Systems*, 2019, pp. 4694–4703.